TAB CONTENT: Biochar for Remediation

Title: Restoring Land. Immobilising Toxins. Securing the Future.

Subtitle: Science-backed biochar solutions for heavy metal immobilisation in Zambian post-mining soils.

The Challenge: Zambia's Mining Legacy

Zambia's economy has long been powered by mining with decades of extraction leaving behind **tailings dams and overburden materials** posing severe environmental and human health risks. Cities like **Kabwe** are severely polluted with Lead (Pb) and Zinc (Zn) from past mining activities, creating toxic hotspots that threaten water tables and communities.

Standard reclamation efforts often fail to address the root issue: the **bioavailability** of these toxins compounded by yearly rainfall. At Green Cycle we don't just cap the problem; we neutralise it!

The Biochar Solution

Our biochar acts as a permanent "lockbox" for heavy metals. Produced through pyrolysis, our carbon-rich material possesses a porous structure and high surface area suited to the diverse chemistry of Zambian mine soils.

Immobilisation Mechanisms:

- **Physical Encapsulation:** The microporous structure traps metal ions like Copper (Cu), Lead (Pb), and Cadmium (Cd), preventing them from leaching into groundwater or runoff into waterways.
- **Ion Exchange:** Our biochar binds with metal cations, effectively removing them from soil solution.
- **pH Modification:** Biochar raises the pH of acidic mine tailings, triggering the hydrolysis of metal ions into stable, insoluble forms

We don't deal in theory. We deal in data. Independent research conducted on **Kabwe's contaminated soils** (one of the world's most notorious lead pollution hotspots) demonstrated the power of biochar:

Links:

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content

Fourteen-year field evidence reveals superior co-benefits of biochar in immobilizing heavy metals and sequestering carbon | Biochar | Springer Nature Link

TAB CONTENT: Microforestry

Title: Microforestry™. The Small Footprint Forest with a Giant Impact.

Subtitle: Functional, fast-growing native forests that bind carbon from day one.

What is Microforestry?

A forest does not need to be a thousand hectares to save the world.

Microforestry is Green Cycle proprietary approach to creating ultra-dense, biodiverse pocket forests in spaces as small as a parking space or a factory perimeter. Derived from the proven **Miyawaki Method**, these are not traditional gardens—they are functioning ecosystems designed to replicate the complex layering of a native Zambian forest .

While the term "Microforestry" is a play on micro pocket forests, the science is serious. We take barren land—be it a corporate campus, a rehabilitated tailings dam, or the edge of a solar farm—and turn it into a self-sustaining **carbon sink** that sequesters carbon from the moment the first sapling touches the soil.

The Green Cycle Difference: Biochar

Most landscaping services stop at planting trees. We start with the planet beneath your feet incorporating **Pyro Biochar** as a mandatory foundation layer.

The Biochar Advantage:

- **Instant Carbon Sequestration:** Every tonne of biochar we add locks away ~2.5 tonnes of CO_2e immediately.
- **Survival Rates:** Biochar-amended soil retains 50% more water and nutrients, boosting sapling survival rates in harsh urban or post-mining environments to over 90%.
- **Heavy Metal Shield:** In post-mining applications, our biochar immobilizes residual heavy metals (Cu, Pb, Zn) in the soil, preventing them from entering the root systems of the native trees we plant .

Target Applications & Urban Solutions

Microforests are the ultimate green infrastructure for Zambia's growing industrial and urban landscape.

1. Industrial & Factory Barriers
Factories produce heat, dust, and noise. A Green Cycle Microforest acts as a **living green wall**. The ultra-dense layering (canopy, tree, sub-tree, shrub) filters airborne

particulate matter, reduces ambient temperatures by up to 10°C via evapotranspiration, and creates a visual buffer .

2. Post-Mining Remediation (The Living Cap)

Tailings dams are often sterile, acidic, or toxic. We utilize "pioneer" native tree species identified by Zambian research (specifically species that tolerate high Copper concentrations) combined with our biochar base to create a **phytostabilization cap**. This stops erosion, prevents acid mine drainage, and reintroduces biodiversity to dead zones .

3. Solar Farm Companionship

Solar panels lose efficiency in high heat. Our Microforests are planted around the perimeter of solar farms to cool the microclimate, reduce dust accumulation on panels, and maximize energy yield.

4. Corporate & Urban Greening

Bring photosynthesis closer to home. Turn sterile corporate parking lots or hospital grounds into "pockets of hope" that staff can use for wellness, shade, and carbon neutrality reporting .

Why the Miyawaki Method?

Traditional tree planting often results in "linear" rows of single species that take decades to mature. The Miyawaki method mimics a natural forest:

- **Density:** 3-5 trees per square meter (not 1 per 10m).
- **Native Only:** We screen local Zambian species (Miombo and cryptosepalum forest variants) to ensure they support local pollinators and wildlife .
- **Speed:** These forests grow faster than conventional plantations. Within 3 years, the forest is maintenance-free. Within 20 years, it achieves the density of an ancient forest .

The Growth Cycle:

- **Year 1:** Intense weeding & watering (establishment of biochar soil matrix).
- **Year 2:** Trees explode in height (competition for light).
- **Year 3:** Canopy closes. No more weeding needed. Forest is self-sustaining.

Benefits Beyond the Leaf

By commissioning a Green Cycle Microforest, you are not just planting trees; you are engineering an asset.

- **Heat Reduction:** Mitigates the "Urban Heat Island" effect, lowering air conditioning costs for adjacent buildings.

- **Stormwater Management:** The dense root systems and biochar act as a sponge, reducing runoff and flash flooding risks in prone areas.
- **Biodiversity:** Microforests reintroduce native birds, bees, and butterflies to industrial zones.
- **ESG & Branding:** A visible, measurable commitment to carbon removal and nature restoration. Unlike offsets bought online, a Microforest is a physical asset on your balance sheet.

TAB CONTENT: Agribusiness

Title: Soil is Infrastructure.

Subtitle: It's time to build—not for a season or two, but permanently.

The Farmer's Most Critical Asset

In agribusiness, we talk about tractors, irrigation pivots, and storage sheds as infrastructure. But the most valuable engineered system on your farm is already beneath your feet.

Soil is living infrastructure! A multi-functional system that supports production, cycles nutrients, manages water, and houses billions of organisms. Yet conventional farming treats soil as a consumable—mining its fertility season after season, then applying synthetic bandages to compensate.

We believe in a different path: **building permanent soil value.**

The Permanent Upgrade

Biochar is not a fertilizer. It is not a compost that disappears after one season. It is a **structural upgrade** to your soil infrastructure—a permanent, recalcitrant carbon matrix that transforms ordinary soil into a high-performance biological system.

Think of biochar as installing **better foundations, better plumbing, and better storage** in your field, all at once.

Here is how biochar upgrades the four critical functions of soil infrastructure:

Advanced "Water Bank" (Hydraulic Infrastructure)

Droughts are becoming more frequent. Loadshedding more frequent, cost of irrigation more expensive. Biochar acts as a microscopic sponge.

- **The Science:** A single gram of biochar can possess a surface area equivalent to a football field. Its porous structure captures and holds moisture, releasing it slowly to plant roots during dry spells.
- **The Benefit:** Reduced irrigation frequency, higher drought resilience, and less water wasted to deep drainage.

"Real Estate for Microbes" (Biological Infrastructure)

Soil biology is the workforce of your farm. But without physical housing, microbes cannot flourish, multiply, or do their job.

- **The Science:** Biochar provides countless microscopic pores that serve as protective habitat for beneficial bacteria, fungi (including mycorrhizae), and protozoa.
- **The Benefit:** Explosive biological activity leads to better nutrient cycling, disease suppression, and root development.

Nutrient "Vault" (Chemical Infrastructure)

Conventional fertilizers are notoriously inefficient. Nitrogen leaches into groundwater. Phosphorus gets locked up. Farmers pay for nutrients that their crops never receive.

- **The Science:** Biochar acts as a cation exchange resin, holding onto positively charged nutrient ions (ammonium, potassium, calcium, magnesium) and releasing them on demand.
- **The Benefit:** Up to 50% reduction in fertilizer leaching. More nutrients stay in the root zone. Less money washes downstream.

Thermal & Structural Stability (Physical Infrastructure)

Compost disappears. Manure mineralizes. But biochar remains.

- **The Science:** Biochar is recalcitrant carbon—stable in soil for hundreds to thousands of years. It does not decompose. It does not compact. It permanently improves soil porosity, aeration, and tilth.
- **The Benefit:** One application. A lifetime of value. You are not renting soil health; you own it.

Our Products: Built for Zambian Agribusiness

Green Cycle Limited offers three biochar product lines, each designed for a specific entry point into your farming system.

Raw / Virgin Biochar (For Fertiliser Blenders)

Description: Pure, activated biochar produced via high-temperature pyrolysis (600°C) from Zambian agricultural residues (maize cobs, rice husks, poultry litter). No inoculants. No amendments. Just performance-grade carbon.

Who It Is For: Fertilizer blending companies, large-scale agribusinesses, and technical farmers who want to incorporate biochar into their existing custom blends or bulk soil programs.

Key Specifications:

- High surface area (>300 m^2/g)
- Alkaline pH (8.5–10.0) ideal for acidic Zambian soils
- Low volatile matter content
- Available in granular or fine mesh

Value Proposition: You control the formulation. We supply the permanent carbon backbone.

Activated & Inoculated Soil-Ready Biochar

Description: Our "plug-and-play" solution. Raw biochar is pre-activated (charged) with a proprietary nutrient solution and inoculated with beneficial microbes, including nitrogen-fixing bacteria and phosphorus-solubilizing fungi. It arrives at your farm ready to work.

Who It Is For: Farmers who want immediate results without the complexity of charging biochar themselves. Ideal for horticulture, vegetables, tobacco, and high-value row crops.

Value Proposition: No lag phase. No guesswork. Apply directly to the furrow or incorporate during land preparation. Your soil biology gets a rocket boost from day one.

Urea Biochar (Pyro Black Urea)

Description: A revolutionary product that solves one of agriculture's most persistent problems: **nitrogen loss**. Conventional urea is notoriously inefficient—up to 50% of applied nitrogen can be lost to volatilization (as ammonia gas) and leaching before crops ever access it.

Green Cycle Limited does not simply coat urea with biochar (as other "black urea" products do). Surface coatings can rub off, degrade, or fail in the field.

Our Innovation: We **mechanically fuse** urea and biochar through a proprietary pelleting process. The urea is not coated on top; it is integrated *into* the biochar matrix. The result is a homogeneous, stable granule where every particle of nitrogen is physically bound or encapsulated in permanent carbon.

Why It Matters:

Feature	Conventional Urea	Coated "Black Urea"	Pyro Black Urea™
Nitrogen retention	Poor (up to 50% loss)	Moderate (coating can fail)	High (mechanical fusion)
Biochar integration	None	Surface coating only	Fused throughout the pellet
Season-to-season value	Zero (urea is spent)	Minimal	**High (biochar remains)**
Mechanism	Dissolves and leaches	Delayed release	Slow release + adsorption

The Critical Advantage:
A bag of conventional urea is fully "spent" within one season. The nitrogen is either taken up by the crop, leached away, or volatilized. No residual value remains.

But with **Pyro Black Urea**, the urea delivers its nitrogen efficiently—and then the biochar stays behind. It remains in your soil permanently, continuing to build your water bank, your microbial real estate, and your nutrient vault for every subsequent season.

One purchase. Immediate nitrogen efficiency. Permanent soil infrastructure.

Who It Is For: Large-scale maize, wheat, soybean, and cotton farmers who rely on urea as their primary nitrogen source but are tired of paying for nitrogen they never use.

Value Proposition: Lower nitrogen costs. Higher use efficiency. And a compounding soil asset that improves your farm's value year after year.

Call to Action

Headline: Stop mining your soil. Start building it.

Text: Whether you are a fertilizer blender looking for bulk raw material, a commercial farmer ready to try Black Urea™, or simply curious how biochar can reduce your input costs—talk to us.

Button: Request a Farm Trial / Get a Quote

SUGGESTED VISUALS FOR THE AGRIBUSINESS TAB

1. **Hero Image:** Split soil profile shot. On the left, compacted, lifeless soil with a stunted root. On the right, dark biochar-rich soil with massive root growth and visible white fungal hyphae.
2. **The "Four Infrastructure" Icons:**
   - Water Bank: A water droplet inside a honeycomb.
   - Microbe Real Estate: A tiny house icon inside a porous sphere.
   - Nutrient Vault: A safe or vault door.
   - Thermal Stability: A thermometer with a shield.
3. **Black Urea Diagram:** A cross-section of a pellet showing urea (yellow) uniformly distributed through black biochar (gray/black), with an arrow labeled "Mechanical Fusion—Not Just a Coating."

This tab now positions Green Cycle Limited not as a "soil amendment seller" but as a **long-term infrastructure partner** to Zambian agribusiness. The Black Urea section specifically addresses a real pain point (urea inefficiency) with a defensible technical moat (mechanical fusion vs. coating).